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### Magnetically coupled galvanically isolated communication using lead frame

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#### Abstract

An integrated circuit package includes a lead frame and an encapsulation that substantially encloses the lead frame. The lead frame further includes a first conductor comprising a first conductive loop and a second conductor galvanically isolated from the first conductor, proximate to and magnetically coupled to the first conductive loop to provide a communication link between the first and second conductor. The second conductor includes a first conductive portion, a second conductive portion, and a wire coupling together the first conductive portion and the second conductive portion.

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## **Background/Summary**

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/029,302, filed on Sep. 23, 2020, which is a continuation of U.S. patent application Ser. No. 16/431,156, filed Jun. 4, 2019, now U.S. Pat. No. 10,833,590, which is a continuation of U.S. patent application Ser. No. 15/943,465, filed Apr. 2, 2018, now U.S. Pat. No. 10,361,632, which is a continuation of U.S. patent application Ser. No. 15/096,515, filed Apr. 12, 2016, now U.S. Pat. No. 10,079,543, which is a continuation of U.S. patent application Ser. No. 14/688,257, filed on Apr. 16, 2015, now U.S. Pat. No. 9,331,004, which is a continuation of U.S. patent application Ser. No. 13/677,088, filed on Nov. 14, 2012, now U.S. Pat. No. 9,035,435. U.S. patent application Ser. No. 17/029,302, U.S. Pat. Nos. 10,833,590, 10,361,632, 10,079,543, 9,331,004, and 9,035,435 are hereby incorporated by reference in their entirety. (2) This application is related to U.S. patent application Ser. No. 13/677,068 of Kung et al., filed Nov. 14, 2012, entitled "Noise Cancellation For A Magnetically Coupled Communication Link Utilizing A Lead Frame," and assigned to the Assignee of the present application. (3) This application is also related to U.S. patent application Ser. No. 13/677,120 of Balakrishnan et al., filed Nov. 14, 2012, entitled "Switch Mode Power Converters Using Magnetically Coupled Galvanically Isolated Lead Frame Communication," and assigned to the Assignee of the present application.

## **BACKGROUND INFORMATION**

### **Field of the Disclosure**

(1) The present invention relates generally to communication between circuits that require galvanic isolation. More specifically, examples of the present invention are related to communication across an isolation barrier in switch mode power converters such as power supplies and power inverters.

### **Background**

(2) Switch mode power converters are widely used for household or industrial appliances that require a regulated direct current (dc) source for their operation, such as for example battery chargers that are commonly used in electronic mobile devices. Off-line ac-dc converters convert a low frequency (e.g., 50 Hz or 60 Hz) high voltage ac (alternating current) input voltage to a required level of dc output voltage. Various types of switch mode power converters are popular because of their well regulated output, high efficiency, and small size along with their safety and protection features. Popular topologies of switch mode power converters include flyback, forward, boost, buck, half bridge and full bridge, among many others including resonant types.

(3) Safety requirements for isolated switch mode power converters generally require the use of high frequency transformers to provide galvanic isolation between the inputs and outputs of the switch mode power converters in addition to the voltage level change at the output.

(4) A major challenge in the market of switch mode power converters is reducing the size and cost of the switch mode power converter while maintaining high performance operating specifications. In known isolated switch mode power converters, the sensing of the outputs of the switch mode power converters and communication of feedback signals for regulating switch mode power converter output parameters such as current or voltage is usually accomplished using external isolation components such as, for example, opto-couplers. These known methods add unwanted additional size as well as cost to switch mode power converters. In addition, opto-couplers are slow

in operation and in many cases limit the feedback bandwidth and the transient response of the switch mode power converter.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.
- (2) FIG. 1A shows the conceptual operation of magnetically coupled conductive loops transmitting and receiving signals to communicate digital or analog information for the purpose of this disclosure
- (3) FIG. 1B illustrates another conceptual example of conductive loops suitable for a bidirectional operation according to teaching of this disclosure.
- (4) FIG. 2A shows an outside view of one example of an integrated circuit package with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.
- (5) FIG. 2B shows an inside view of one example structure of galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the example integrated circuit package of FIG. 2A in accordance with the teachings of the present invention.
- (6) FIG. 2C shows an outside view of one example of an integrated circuit package with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.
- (7) FIG. 2D shows an inside view of one example structure of galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the example integrated circuit package of FIG. 2C in accordance with the teachings of the present invention.
- (8) FIG. 3A shows an outside view of an example of an integrated circuit package with a magnetically coupled communication link formed by isolated conductors of the lead frame inside the encapsulated portion the integrated circuit package in accordance with the teachings of the present invention.
- (9) FIG. 3B shows an inside view of one example of a lead frame inside an integrated circuit package with a magnetically coupled communication link formed by magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.
- (10) FIG. 3C shows an inside view of an example of a lead frame of a multi-die isolated controller integrated circuit package with a magnetically coupled communication link between the controller dice, the link being formed by magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.
- (11) FIG. 3D shows an example side-view of a jumper bond wire coupled to an integrated circuit die and a conductive loop of an isolated conductor of a lead frame inside an integrated circuit package in accordance with the teachings of the present invention.
- (12) FIG. 4A shows a tilted 3D (3 dimensional) view of an inside view of one example of a lead frame of an integrated circuit package with a magnetically coupled communication link formed by magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.



(13) FIG. 4B shows a tilted 3D view of one example of a lead frame of a multi-die isolated controller integrated circuit package with a communication link between the controller dice formed by magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

(14) FIG. 5 shows a schematic of one example of a synchronous flyback switch mode power converter with secondary control utilizing one example of a multi-die isolated controller integrated circuit package with a magnetically coupled communication link between the controller dice formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

(15) FIG. 6 shows a schematic of one example of a flyback switch mode power converter utilizing one example of a multi-die isolated controller integrated circuit package including a bidirectional magnetically coupled communication link between the controller dice inside the encapsulated portion of the integrated circuit package in which output information is transferred to a primary side through the magnetically coupled communication link and a line zero-cross detection signal is transferred to the secondary side through the magnetically coupled communication link in accordance with the teachings of the present invention.

(16) Corresponding reference characters indicate corresponding components throughout the several views of the drawings. Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention.

#### DETAILED DESCRIPTION

(17) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present invention. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present invention.

(18) Reference throughout this specification to “one embodiment”, “an embodiment”, “one example” or “an example” means that a particular feature, structure or characteristic described in connection with the embodiment or example is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment”, “in an embodiment”, “one example” or “an example” in various places throughout this specification are not necessarily all referring to the same embodiment or example. Furthermore, the particular features, structures or characteristics may be combined in any suitable combinations and/or subcombinations in one or more embodiments or examples. Particular features, structures or characteristics may be included in an integrated circuit, an electronic circuit, a combinational logic circuit, or other suitable components that provide the described functionality. In addition, it is appreciated that the figures provided herewith are for explanation purposes to persons ordinarily skilled in the art and that the drawings are not necessarily drawn to scale.

(19) In some applications multiple controllers may be housed in a single integrated circuit package. Each controller is fabricated as a semiconductor die. The present application discloses an integrated circuit package structure that enables communication between the controllers with galvanic isolation using magnetic coupling between portions of the lead frame while adding little or no cost to the overall package.

(20) An integrated circuit package typically includes a lead frame. The lead frame provides mechanical support for a single die or for multiple dice that may be housed within the integrated

circuit package. In general, the lead frame typically includes a die attach pad to which a semiconductor die may be attached. In addition, the lead frame generally also includes leads that serve as electrical connections to circuits external to the integrated circuit package. The lead frame is generally constructed from a flat sheet of metal. The flat sheet of metal may be stamped, etched, punched, etc., with a pattern, which defines the die attach pads and various leads of the lead frame. (21) As mentioned above, isolation is often provided in known switch mode power converters using external isolation components such as for example opto-couplers or through the use of an extra bias (e.g., feedback) winding on the transformer core that is magnetically coupled to the secondary winding. These known methods add unwanted additional size as well as overall cost to switch mode power converters. Isolation is provided in examples in accordance with the teachings of the present invention with magnetically coupled conductive loops formed by galvanically isolated conductors of the lead frame inside the encapsulated portion of an integrated circuit package structure, which provides a magnetically coupled communication link between isolated circuits. In various examples, the isolation provided by the magnetically coupled communication link formed by isolated conductors of the lead frame of the integrated circuit package in accordance with the teachings of the present invention may be utilized in a variety of applications including switch mode power converters that require galvanic isolation between the primary and secondary sides of the switch mode power converters. Some example switch mode power converters utilizing an integrated circuit package having a magnetically coupled communication link formed by isolated conductors of the lead frame of the integrated circuit package include, but are not limited to, synchronous flyback, isolated flyback, isolated synchronous flyback, buck, forward, half-bridge and full-bridge topologies in accordance with the teachings of the present invention.

(22) For the purpose of this disclosure, a physical closed path for current is referred to as a loop. A loop may include different elements such as conductors (that in examples of this disclosure could be formed by lead frame and bond wires inside an IC package) as well as electrical components that are in path of the circulating current. Each element in the loop forms a part of the loop, and combination of one or more elements in the loop is referred to as a partial loop. In the context of magnetic field coupling, a loop enclosing a magnetic field is typically referred to as having one or more turns. Each turn corresponds to one enclosure of the magnetic field.

(23) FIGS. 1A and 1B show the conceptual operation of magnetically coupled conductive loops transmitting and receiving signals to communicate operational information for example in a controller IC of a switch mode power converter in accordance with the teachings of the present invention. In FIG. 1A the magnetically coupled communication link **100** includes an outer conductive loop **102** coupled to a transmit circuit **110** and an inner conductive loop **105** coupled to a receive circuit **130**. The outer conductive loop **102** in one example includes a pulse current source **114**, injecting a pulse current **120** to conductive loop **102**. In embodiments, the transmit circuit **110** may communicate information utilizing the transmitter current  $I_{sub.T}$  **120**. In one example, circuits within transmit circuit **110** may control various properties of the transmitter current  $I_{sub.T}$  **120** to communicate information to the receive circuit **130**. When the transmitter current  $I_{sub.T}$  **120** is changing or varying in magnitude over time, it produces a changing magnetic field in the proximity of the conductor of the inner conductive loop **105**. Due to the laws of electromagnetic induction, a voltage is generated across a conductor that is subjected to a changing magnetic field. The pulse current  $I_{sub.T}$  **120** in one example has a time when it is rising, a time when it is falling and an amplitude. The changing flux generated by outer conductive loop **102** due to transmitter current  $I_{sub.T}$  **120** has a direction entering the surface of the page. Marker **108** illustrates the overall magnetic field that passes through both transmitter loop **102** and receiver loop **105**. In general, the “X” symbol as illustrated for marker **108** denotes magnetic field or flux into the page, while a dot symbol for a marker symbol denotes magnetic field or flux out from the page.

(24) In the embodiment therefore, receiver voltage  $V_{sub.R}$  **140** is induced due to the changing magnetic field generated by changes in current  $I_{sub.T}$  **120** and may result in receiver current

I.sub.R **138** in the direction illustrated in FIG. **1A**.

(25) The receive circuit **130** may include circuits which may receive the voltage and/or current induced by the transmit circuit **110** and interprets the voltage and/or current as information. Properties of the transmitter current I.sub.T **120** which may be controlled to communicate information may include the magnitude and rate of change of the transmitter current I.sub.T **120**. In the example of depicted transmitter current I.sub.T **120** the rising and falling slopes defined by the pulse waveform **120** induce pulsating voltage V.sub.R **140** with a positive amplitude during rising time and a negative amplitude during falling time of the transmitter current pulse waveform **120**. The receive circuit **130** in one example may include a comparator **132** responding to a comparison of the amplitude of induced voltage pulses V.sub.R **140** of receive circuit **130** to a threshold voltage V.sub.Th **134**.

(26) The communicated signals may take the form of digital information or of analog information. In the case of digital information, communication can be in the form of binary signals or more complex encoded digital data as will be known to one skilled in the art. It is appreciated that other communication techniques may be used. In other examples, communication techniques which take advantage of the relationship between the transmitter current I.sub.T **120** and the resultant induced receiver voltage V.sub.R **140** and receiver current I.sub.R **138** received by the receive circuit **130** may be utilized.

(27) FIG. **1B** illustrates another example of the magnetically coupled communication link **150**. In one example communication link **150** could be suited for bidirectional communication and includes two conductive loops. First loop **152** and second loop **155** are positioned to enclose the maximum common magnetic field area. In contrast to the example of FIG. **1A**, that could be better suited to a unidirectional communication, loops **152** and **154** of bidirectional example of FIG. **1B** have approximately the same dimensions. For the best bidirectional operation, physical symmetry of the loops is important resulting in approximately equal bidirectional behavior. The magnetic field or flux in the first loop **152** and second loop **155** has a direction into the page.

(28) The operational/functional difference between FIG. **1A** and FIG. **1B** is that in FIG. **1B** both first loop **152** and second loop **155** are coupled to the transceiver (transmit/receive) circuits **160** and **180** respectively. Transceiver circuit **1**, **160** through the selection switch S1 **163** may couple either a transmit circuit **162** or receive circuit **165** to the first loop **152**.

(29) Transceiver circuit **2**, **180** through the selection switch S2 **183** may couple either a transmit circuit **182** or receive circuit **185** to the second loop **155**.

(30) If the transceiver circuit **1** **160** is coupled as a transmit circuit to inject a current pulse I.sub.TR1 **170** to the first loop, then transceiver circuit **2** **180** through the second loop **155** and switch S2 **183** would be coupled as a receive circuit to receive the communicated signal as a current pulse I.sub.TR2 **190** or as a voltage pulse V.sub.TR2 **187**.

(31) On the other hand if the transceiver circuit **2** **180** is coupled as a transmit circuit to inject a current pulse I.sub.TR2 **190** to the second loop, then transceiver circuit **1** **160** through the first loop **152** and by the controlled function of the switch S1 **163** would be coupled as a receive circuit to receive the communicated signal as a current pulse I.sub.TR1 **160** or as a voltage pulse V.sub.TR1 **167**.

(32) The transmit circuits **162** and **182** in the transceiver circuits **160** and **180** could include pulse current sources **164** and **184** respectively and the receive circuits **165** and **185** in the transceiver circuits **160** and **180** could include comparator circuits **166** and **186** with threshold voltages **168** and **188** respectively.

(33) To illustrate an example of practical application in IC industry, FIG. **2A** and FIG. **2B** show an example integrated circuit package **210** with galvanically isolated magnetically coupled conductive loops formed by galvanically isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. In example illustrated in FIGS. **2A** and **2B**, there are external pins **201**, **202**, **203** and **204**, as well as

external pins **205**, **206**, **207** and **208**, on two sides of integrated circuit package **210**. In the example, all of the external pins are part of the lead frame **298** that comprises the internal conductive elements **296** and **297** that are fundamentally part of integrated circuit package **210** before any bond wires, such as bond wires **295**, are introduced, and extend from an encapsulation **299** of integrated circuit package **210** as shown. In one example, lead frame **298** may be comprised of known conductive materials utilized for lead frames in integrated circuit packaging, such as for example copper, and is substantially flat and embedded in a molding compound of integrated circuit package **210**. In the example, lead frame **298** provides electrical connectivity to and from circuitry coupled to pins **201** to **208** of package **210** as well as provides mechanical support for the connection of bond wires **295**.

(34) For the purposes of this disclosure, an “encapsulation” of an integrated circuit package may be considered to be any external body, encasing or molding that surrounds or encloses a portion of the lead frame which may include one or more integrated circuit dice disposed therein, as well as connections from the integrated circuit die pads to the lead frame and pins of the integrated circuit package. An example encapsulation may be made from molded non-ferrous insulating material, plastic, ceramic covers or the like. In some examples, the encapsulation of the integrated circuit package may or may not provide hermetic sealing to protect the items encased therein from external elements.

(35) For the purposes of this disclosure, the term “integrated circuit package” refers to the type of packages used generally for integrated circuits. It is appreciated that some embodiments of this invention may have no integrated circuits in the package such as the examples in FIGS. 2A, 2B, 2C and 2D.

(36) FIG. 2B shows a view inside the encapsulation **299** revealing one example structure of the galvanically isolated magnetically coupled conductive loops **215** and **211** formed by isolated first and second conductors **296** and **297** of lead frame **298** of the example integrated circuit package **210** of FIG. 2A in accordance with the teachings of the present invention. In particular, as shown in the illustrated example, lead frame **298** includes first conductor **297** and a second conductor **296**, which are encapsulated in insulating molding compound material within encapsulation **299**. In one example, first and second conductors **297** and **296** of lead frame **298** may be formed from a flat sheet of metal by etching, stamping, punching, or the like, to form first conductive partial loop **215** in first conductor **297**, and a second conductive partial loop **211** in second conductor **296**. In the example depicted in FIG. 2B, a bond wire **295** is coupled to second conductor **296** as shown to couple together portions of second conductive partial loop **211**. In the depicted example, second conductor **296** is galvanically isolated from first conductor **297**. In one example, bond wire **295** has a sufficient path length to provide sufficient isolation space from first conductor **297** in order to maintain the galvanic isolation between first conductor **297** and second conductor **296**. In another example not illustrated, it is appreciated that one or more additional bond wires may be included coupling together portions of first conductive loop **215** and/or second conductive loop **211**. It is appreciated that circuit elements connected between pins **201**, **202** and pins **203**, **204** are needed to complete the partial conductive loop **215** shown in order to transmit or receive signals through the communication link. Likewise it is appreciated that circuit elements connected between pins **205**, **206** and pins **207**, **208** are needed to truly complete the partial conductive loop **211** shown in order to transmit or receive signals through the communication link. However for the purposes of this description, partial conductive loops **211** and **215** may be referred to as conductive loops. It is appreciated that this comment also extends to the subsequent discussion of FIG. 1C and FIG. 1D below.

(37) As shown in the example, second conductive loop **211** is disposed within encapsulation **299** proximate to and magnetically coupled to a first conductive loop **215** to provide a communication link between the galvanically isolated first conductor **297** and second conductor **296** in accordance with the teachings of the present invention. In one example, magnetically coupled portions of the

first and second conductive loops **215** and **211** are substantially flat and disposed substantially in a same plane. As shown in the illustrated example, the first and second conductive loops **215** and **211** each consist of one turn. In one example, the communication link provided by the magnetic coupling between second conductive loop **211** and first conductive loop **215** is utilized to communicate one or more signals between galvanically isolated second conductor **296** and first conductor **297** of the lead frame **298** in accordance with the teachings of the present invention. In one example, a transmitting signal is applied between first terminal T1 pins **205** and **206**, and second terminal T2 pins **207** and **208**. It is sometimes desirable to have more than one external pin common to a terminal for convenience of assembly on a circuit board.

(38) Continuing with the illustrated example, the signal is received by first conductive loop **215** through the magnetic coupling from second conductive loop **211** between first terminal R1 pins **201** and **202**, and second terminal R2 pins **203** and **204**. In another example, it is appreciated that the signal can also be communicated in the opposite direction to provide bidirectional communications.

(39) As shown in the example, first terminal R1 pins **201** and **202** are coupled to first conductive loop **215** through a lead frame connection **216** and second terminal R2 pins **203** and **204** are coupled to first conductive loop **215** through a lead frame connection **218**. In the example, the signals at terminals T1 and R1 are in phase in accordance with magnetic coupling and induction laws. In one example, the connections of each terminal T1 **214**, T2 **212**, R1 **216** and R2 **218** to the respective pairs of outside pins **205/206**, **207/208**, **201/202** and **203/204**, as described above, by providing multiple assembly options simplifies the physical connections on a circuit board on which integrated circuit package **210** is mounted.

(40) FIG. 2C shows an outside view of one example of an integrated circuit package **250** with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame **298** inside the encapsulated portion of the integrated circuit package **250** in accordance with the teachings of the present invention. It is appreciated that integrated circuit package **250** of FIG. 2C shares many similarities with integrated circuit package **210** of FIG. 2A. For instance, integrated circuit package **250** of FIG. 2C includes an encapsulation **299** in which a lead frame **298** is disposed. However, one difference is that instead of having external pins arranged at two sides of the integrated circuit package, integrated circuit package **250** includes external pins **252**, **254**, **256** and **258** arranged on one side of integrated circuit package **250**. In the example, all of the external pins are part of the lead frame **298** of integrated circuit package **250** and extend from a single side of the encapsulation **299** of integrated circuit package **250** as shown.

(41) FIG. 2D shows a view inside the encapsulation **299** of one example structure of galvanically isolated magnetically coupled conductive loops **215** and **211** formed by the isolated first and second conductors **297** and **296** of the lead frame **298** of the example integrated circuit package of FIG. 2C in accordance with the teachings of the present invention. It is appreciated that the view inside the encapsulation **299** of integrated circuit package **250** shares many similarities with the view inside the encapsulation **299** of integrated circuit package **210**. For instance, as shown in FIG. 1D, lead frame **298** includes first conductor **297** and a second conductor **296** encapsulated in insulating material within encapsulation **299**. In the depicted example, second conductor **296** is galvanically isolated from the first conductor **297**. As shown in the example, a second conductive loop **260** of second conductor **298** is disposed within encapsulation **298** proximate to and magnetically coupled to a first conductive loop **255** included in first conductor **297** to provide a communication link between the galvanically isolated first conductor **297** and second conductor **296** in accordance with the teachings of the present invention. One difference from the example illustrated in FIG. 2B is that in the example illustrated in FIG. 2D, there is no bond wire **295** included in first conductive loop **255** and/or second conductive loop **260**.

(42) In the example illustrated in FIG. 2D, the communication link provided by the magnetic coupling between second conductive loop **260** and first conductive loop **255** is utilized to

communicate one or more signals between galvanically isolated second conductor **296** and first conductor **297** of the lead frame **298** in accordance with the teachings of the present invention. In the example, the transmitting signal is applied between first terminal **T1** pin **258** and second terminal **T2** pin **252**. As shown in the example, first terminal **T1** pin **258** and second terminal **T2** **252** are coupled to second conductive loop **260**. The signal is received by first conductive loop **255** through the magnetic coupling from second conductive loop **260** between first terminal **R1** pin **256** and second terminal **R2** pin **254**. In another example, it is appreciated that the signal can also be communicated in the opposite direction to provide bidirectional communications.

(43) FIG. **3A** shows an outside view of an example of an integrated circuit package **315** with a magnetically coupled communication link formed by isolated conductive loops of the lead frame **398** inside the encapsulation **399** of the integrated circuit package **315** in accordance with the teachings of the present invention. In the example illustrated in FIG. **3A**, there are external pins **301, 302, 303, 304, 305, 306, 307, 309, 310, 311** and **312** as shown. In the example, all of the external pins are part of the lead frame **398** of integrated circuit package **315** and extend from the encapsulation **399** of integrated circuit package **315** as shown. In one example, lead frame **398** may be comprised of known conductive materials utilized for lead frames in integrated circuit packaging, such as for example copper, and is substantially flat and encapsulated in a molding compound. In the example, lead frame **398** provides electrical connectivity to and from internal circuitry within encapsulated portion of the integrated circuit package **315** as well as provides mechanical support for integrated circuits and bond wires inside package **315**.

(44) FIG. **3B** shows a view inside the encapsulation **399** revealing one example the structure of galvanically isolated magnetically coupled conductive loops **337** and **335** formed by isolated first and second conductors **397** and **396** of the lead frame **398** of the example multi-die isolated controller integrated circuit package **315** of FIG. **3A** in accordance with the teachings of the present invention. In particular, as shown in the illustrated example, lead frame **398** includes first conductor **397** and a second conductor **396** encapsulated in insulating material within encapsulation **399**. As shown in the depicted example, a first conductor **397** includes a first conductive loop **337** and second conductor **396** includes a second conductive loop **335**. As shown in the example, second conductive loop **335** is disposed within encapsulation **399** proximate to and magnetically coupled to a first conductive loop **337** to provide a communication link between the galvanically isolated first conductor **397** and second conductor **396** in accordance with the teachings of the present invention. In one example, first conductor **397** also includes an optional third conductive loop **338**, which in one example may be utilized for noise cancellation and is attached to tie bar **339** as shown. In one example, tie-bar **339** provides a mechanical support connection during the manufacture of package **315** before the lead frame **398** is encapsulated with encapsulation **399**. In one example the encapsulation **399** is injection molded with a molding compound. The communication link provided by the magnetic coupling between second conductive loop **335** and first conductive loop **337** is utilized to communicate one or more signals between the galvanically isolated second conductor **396** and first conductor **397** of the lead frame **398** in accordance with the teachings of the present invention.

(45) In FIG. **3B** the current signal from transmit circuit **367** that in one example is included in controller die **360**, assembled on the die pad **334**, is injected through the bond wire **344** from node **341** of transmit circuit **367**. The current signal flows to the end node **343** of the bond wire **344** and then completes the second loop **335**, flowing through the lead frame back to the transmit circuit **367** through the bond wire **361**. The injected signal generates a changing magnetic field that induces a voltage signal in the first conductive loop **337** and results in a current signal closing from the first conductive loop **338** to the receive circuit **335** through the bond wires **377** and **357**. The receive circuit **335** may be included in the first controller die **355** assembled on the die pad **333** that is the primary ground.

(46) FIG. **3C** shows another view inside the encapsulation **399** in which a first isolated control die

**350** is mounted on and coupled to the first conductor **397** and a second isolated control die **360** is mounted on and coupled to the second conductor **396** in accordance with the teachings of the present invention. In the illustrated example, first isolated control die **350** is mounted on die pad **333** and second isolated control die **360** is mounted on die pad **334** as shown. In the illustrated example, die pads **333** and **334** are utilized as isolated primary and secondary ground pads, respectively. In the example shown in FIG. 3C, a magnetically coupled communication link between the first isolated control die **350** and second isolated control die **360** is formed by the magnetically coupled communication link between the first conductive loop **337** and second conductive loop **335** in accordance with the teachings of the present invention. In one example, multi-die isolated controller integrated circuit package **315** may be utilized in a switch mode power converter such as for example a synchronous flyback switch mode power converter with secondary control in accordance with the teachings of the present invention.

(47) Products and applications that require low output voltages, such as for example 5V and below, in some cases, use synchronous rectification to achieve high efficiency and compact form factor. Synchronous rectification utilizes a MOSFET (metal oxide semiconductor field effect transistor) that is switched to behave like a rectifier, in place of an output rectifier diode, to reduce voltage drop and power loss. The switching action of an output MOSFET rectifier is synchronized with the main power switch with well-controlled gating signals. In one example, first isolated control die **350** includes a primary control circuit and a switch (in one example a MOSFET) for use in the primary side of a synchronous flyback switch mode power converter, and the second isolated control die **369** includes a secondary control circuit for use in the secondary side of the synchronous flyback switch mode power converter. In various examples, the primary control circuit and switch/MOSFET may be implemented with a monolithic or hybrid structure for the first isolated control die **350**.

(48) As shown in the example illustrated in FIG. 3C, the primary switch (or MOSFET) is included in first isolated control die **350**. In one example, the drain terminal D **340** of the MOSFET is coupled through bond wires **342** to pin **301**. The source terminal S **345** of the MOSFET is coupled through bond wires **347** to the primary ground die pad **333**, which is accessible through source pin **302**. In the illustrated example, there is a wide clearance (i.e., missing pins), often referred to as creepage distance, between drain pin **301** and source pin **302**. In the illustrated example, the wide pad of source pin **302** is internally coupled to a primary ground pad **333**, which may also serve as a heat sink. In one example, pins **303** and **304** are coupled to first isolated control die **350** through bond wires **352** and **354**, respectively, to connect the first isolated control die **350** to external circuitry such as for example line under voltage (example of UV **536** in FIG. 5) and supply bypass capacitor (example of BP **531** in FIG. 5).

(49) Bond wire **336** couples the third conductive loop **338** to the first isolated control die **350**. Due to a changing magnetic field generated by a changing current flowing in second conductive loop **335**, a voltage signal is induced in the first conductive loop **337**. In the example of FIG. 3C first conductive loop **337** is coupled to the third conductive loop **338** (the optional noise cancellation loop that is an extension of the first conductive loop). The induced voltage signal is coupled through bond wires **336** and **347** to the receive circuit that in one example is included in the primary die **350** on the primary ground die pad **333**.

(50) In the example, pin **305** is attached to second conductive loop **335** of second conductor **396** for mechanical support. The signal communicated from second isolated control die **360** is coupled to second conductive loop **335** through bond wires **344** and **361**, which complete the second conductive loop **335**. As shown in the example, bond wire **344** is a connection coupling second conductive loop **335** at point **343** to second isolated control die **360** at point **341**. Pin **312** in one example is coupled through the current sense bond wire **371** to the secondary Ground pad **334** and the sensed voltage drop on bond wire **371** is coupled to second isolated control die **360** through bond wires **370** and **372** and is utilized for a secondary current measurement. In one example, the

bond wires **362**, **364**, **365** and **366** are coupled between second isolated control die **360** and pins **306**, **307**, **308**, **309** and **310**, respectively, and are utilized for the input/output of secondary signals. In one example, pin **311** provides access to secondary ground pad **334** as shown.

(51) In one example, the slot on secondary ground pad **334** under the second isolated control die **360** makes the second conductive loop **335** longer by forcing the current through the second conductive loop **335** to ground pad **334** to flow closer and parallel to the first conductive loop **337** to improve magnetic coupling. The smaller first conductive loop **337** proximate to and surrounded by the second conductive loop **335** provides a strong magnetic coupling of first and second conductive loops in accordance with the teachings of the present invention. In one example, lead frame **398** is flat, but in other examples some portions of the lead frame **398** may be up set and/or down set for optimum vertical positioning to accommodate die thickness, optimizing bond wire profiles and to align to tie bars and external pins of the integrated circuit package **315**.

(52) FIG. 3D shows an example side-view of a bond wire **336**, which as mentioned above is an electrical connection, and is coupled to second isolated control die **360** at point **341** and second conductive loop **335** at point **343** of the second conductive loop **335** in accordance with the teachings of the present invention. As shown in the example, bond wire **336** is at a higher level than the level of second conductive loop **335** and pin pad **334** of lead frame **398**. As shown, bond wire **336** has sufficient span to complete second conductive transmitter loop **335** and to be isolated from the first conductive loop **337**.

(53) FIG. 4A shows a tilted 3D (3 dimensional) view of an inside view of one example of a lead frame of an integrated circuit package with a magnetically coupled communication link that is formed with magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. FIG. 4A shows the illustrated lead frame structure shares similarities with the lead frame **398** structures of FIG. 3B and FIG. 3C. In particular, in the example of FIG. 4A, the lead frame structure includes a first conductor including a primary die pad **433** and a first conductive loop **437**, as well as a third conductive loop **438**, which correspond to die pad **333**, first conductive loop **337**, as well as third conductive loop **338**, respectively, of FIGS. 3B and 3C. In addition, in the example of FIG. 4A, the lead frame structure also includes a second conductor including a secondary die pad **434** and a second conductive loop **435**, which correspond to die pad **334** and second conductive loop **335**, respectively, of FIGS. 3B and 3C. In FIGS. 4A and 4B the tie-bar connection **439A** to support the third conductive loop **438** is at a different location than the tie-bar connection **339** to support the third conductive loop **338** in FIGS. 3B and 3C and the tie-bar connection **439B** of FIGS. 4A and 4B is not present in the lead frame **398** of FIGS. 3B and 3C. Consequently, the lead frame design shown in FIGS. 3A, 3B and 3C has no tie bar connections on the top and bottom sides of the encapsulation increasing the external creepage distance between the primary and secondary conductors of the lead frame to the shortest distance between external pins **304** and **305** or between external pins **301** and **312**, whichever is smaller, measured along the external surface of the encapsulation.

(54) FIG. 4B shows a tilted 3D view of another inside the encapsulation view of one example of a lead frame of a multi-die isolated controller integrated circuit package with a communication link between the controller dice that are formed with magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. FIG. 4B shows a lead frame structure sharing similarities with the lead frame structure shown in FIG. 4A. In the example illustrated in FIG. 4B, the primary and secondary dice are shown mounted on the lead frame. As shown in the illustrated example, the primary switch **450** and controller **451** are on different dice—which is commonly referred to as a hybrid structure. In the example of FIG. 4B, the power MOSFET has a high power rating, which results in a separate die **450** having a large size that covers substantially all of the primary ground die pad **433**. In the illustrated example, the primary control die **451** is



mounted over part of the third conductive loop **438** as shown. The secondary control die **452** is mounted on the secondary ground die pad **434** as shown.

(55) It is appreciated that an integrated circuit package having a magnetically coupled communication link between galvanically isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention may be utilized in a variety of different applications. Although several different switch mode power converter topologies utilizing such an integrated circuit package having a magnetically coupled communication link are described herein, it is appreciated that the specific examples described in this disclosure are provided for explanation purposes, and that other applications may utilize a magnetically coupled communication link between galvanically isolated conductors of a lead frame inside the encapsulated portion of an integrated circuit package in accordance with the teachings of the present invention.

(56) To illustrate, FIG. 5 shows one such example application with a schematic of an example synchronous flyback switch mode power converter **500** with secondary control utilizing one example of a multi-die isolated controller integrated circuit package **560** having a magnetically coupled communication link **540** between the controller dice that is formed with galvanically isolated conductors of a lead frame inside the encapsulated portion of the integrated circuit package **560** in accordance with the teachings of the present invention.

(57) It is appreciated that secondary control for a flyback converter has advantages of tighter output regulation and faster response to load transients. However, as discussed previously, conventional methods of secondary control often use external isolation devices, such as for example opto-couplers, which increase the complexity and cost of the switch mode power converter. By using an example multi-die isolated controller integrated circuit package **560** having a magnetically coupled communication link **540** with isolated primary and secondary control dice, externally added isolation components such as opto-couplers are no longer needed in accordance with the teachings of the present invention. Furthermore, since integrated circuit package **560** provides a magnetically coupled communication link by using the lead frame of the integrated circuit package as discussed previously, galvanic isolation is maintained between the primary and secondary sides of the switch mode power converter at nearly zero additional cost, without having to add external isolation components in accordance with the teachings of the present invention.

(58) In the example synchronous flyback switch mode power converter **500**, the primary and secondary controllers are galvanically isolated from one another, but there is still reliable communication between the primary and secondary controllers. It is appreciated that although the example of FIG. 5 shows a synchronous flyback converter, a standard flyback converter, where synchronous MOSFET **550** is replaced by a diode, would also benefit from the teachings of the present invention.

(59) In the example illustrated in FIG. 5, synchronous flyback switch mode power converter **500** includes an input coupled to an ac line **505** as shown. A full-bridge rectifier **510** is coupled to ac line **505** to generate rectified ac **515**, which is filtered by capacitance C.sub.F **517**. The rectified ac **515** is coupled to be received by energy transfer element **520**, which includes a primary winding **521** and a secondary winding **522** as shown. In the illustrated example, clamp circuit **525** is coupled across primary winding **521** of energy transfer element **520** as shown.

(60) In the depicted example, a switching device S1 **530** is coupled to the input of synchronous flyback switch mode power converter **500** at the primary ground **501** and to the energy transfer element **520** at primary winding **521**. In the illustrated example, switching device S1 **530** may be included in a monolithic or hybrid structure in the integrated circuit package **560**. As shown in the depicted example, switching device S1 is controlled by control signal **539** from the primary controller die **535** and regulates the energy transfer through primary winding **521** of transformer **520** to the secondary winding **522** in response to line and load changes. Clamp circuit **525**, which in the illustrated example is a diode-resistor-capacitor circuit, is coupled to clamp the turn-off

spikes that result from the leakage inductance from primary winding **521** across the switching device **S1 530**.

(61) As shown in the example of FIG. 5, switch **S2 550** and anti-parallel diode **D2 555** are coupled to secondary winding **522** at the secondary side and serve as a synchronous rectifier of synchronous flyback switch mode power converter **500**. In one example, the diode **D2 555** is an externally connected Schottky diode. In one example, switch **S2 550** is controlled by a signal from the SR pin of the secondary controller die **565**. Whenever the voltage at SR terminal **575** rises to a value higher than the gate threshold voltage, the synchronous rectifier provided by switch **S2 550** begins conducting current. The secondary ripple is smoothed by output filter capacitance **C1 586** and the dc output voltage **Vo 580** is applied to load **585** with load current **Io 582**. The output voltage **Vo 580** is sensed through the resistor divider comprised of resistors **572** and **574**, which is coupled to the feedback pin **FB 573** of the secondary controller. It is appreciated that in other examples resistors **574** and **572** could be integrated within integrated circuit **560** while still benefiting from the teachings of the present invention.

(62) At startup, primary die **535**, which is referenced to the primary ground **501**, starts the switching of switch **S1 530**, which starts the transfer of energy to the secondary side. The bypass pin **BP 531** is externally coupled to the bypass capacitor **532**. The line under voltage pin **UV 536** is externally coupled through resistor **537** to the ac input line, which in another example could be coupled to a rectified ac bus **515**. Communication between the primary die **535** and secondary die **565** is through a magnetic coupling provided by a magnetically coupled communication link **540** formed by isolated conductors of the lead frame of the integrated circuit package in accordance with the teachings of the present invention. In various examples, the communication link **540** is implemented using galvanically isolated conductive loops included in the lead frame of the integrated circuit package as described above in accordance with the teachings of the present invention.

(63) FIG. 6 shows a schematic of one example of a flyback switch mode power converter **600** utilizing one example of a multi-die isolated controller integrated circuit package including a bidirectional magnetically coupled communication link between the controller dice inside the encapsulated portion of the integrated circuit package in which output information is transferred to a primary side through the magnetically coupled communication link and a line zero-cross detection signal is transferred to the secondary side through the magnetically coupled communication link in accordance with the teachings of the present invention.

(64) In the example illustrated in FIG. 6, flyback switch mode power converter **600** includes an input coupled to an ac line **605** as shown. A full-bridge rectifier **610** is coupled to ac line **605** to generate rectified ac **615**, which is filtered by capacitance **C.sub.F 617**. The rectified ac **615** is coupled to be received by energy transfer element **620**, which includes a primary winding **621** and a secondary winding **622** as shown. In the illustrated example, clamp circuit **625** is coupled across primary winding **621** of energy transfer element **620** as shown.

(65) In the depicted example, a switching device **S1 630** is included in an integrated circuit package **660**. In one example, the switch die and the primary control die may be structured as monolithic or hybrid dice. In the example, switching device **S1 630** is coupled to the input of flyback switch mode power converter **600** at the primary ground **601** and to the energy transfer element **620** at primary winding **621**. As shown in the depicted example, switching device **S1 630** is controlled by control signal **639** from the primary controller die **635** and regulates the energy transfer through primary winding **621** of transformer **620** to the secondary winding **622** in response to line and load changes. Clamp circuit **625**, which in the illustrated example is a diode-resistor-capacitor circuit, is coupled to clamp the turn-off spikes that result from the leakage inductance from primary winding **621** across the switching device **S1 630**. In the illustrated example, the secondary rectifier diode **D2 655** in flyback only conducts current during an off-time of the primary switch **630**.

(66) The secondary ripple is filtered by the output filter capacitance **C1 686** and the dc output

voltage **Vo 680** is applied to the load **685** with load current **Io 682**. The output voltage **Vo 680** is sensed through resistor divider comprised of resistors **672** and **674**, which is coupled to the feedback pin **FB 673** of the secondary controller die **665** and is referenced to secondary ground **691** isolated from the primary ground **601**. In one example, feedback signal **673** is a data signal that is transmitted through the magnetic coupling of the lead-frame communication loop **641** and received by the primary die **635** in reference to the primary ground **601**. In one example, the **FB signal 673**, transferred by lead frame magnetic coupling of the communication link **640** to the primary side controller die **635**, may be either a digital or an analog signal. **FB signal** is utilized in combination with the input line information received at pin **634** through resistor **637** to generate gate control signal **639** to control the switching of switch **S1 630** to regulate the transfer of energy through energy transfer element **620** to the output. In one example, lead frame communication link **640** includes unidirectional communication links **641** and **642** to transmit one or more control signals between dice **665** and **635** in accordance with the teachings of the present invention. In another example, lead frame communication link **640** includes a single bidirectional communication link (as depicted in FIG. 1B) using the same magnetically coupled lead frame loop to transmit one or more control signals in either direction between dice **665** and **635** in accordance with the teachings of the present invention.

(67) In one example, the specific control function of the example flyback switch mode power converter **600** of FIG. 6 also utilizes a zero-cross signal of the ac line that is sensed at ac line input **605** through the shunt connected resistors **602** and **603** at the common point **604** referenced to the primary ground **601** as shown. In the example, zero sense signal **606** is coupled to primary die **635** and referenced to primary ground **601**, and is transmitted through the magnetic coupling of the lead-frame communication loop **642** and received by the secondary die **665** with reference to the secondary ground **691**, which could be utilized as an isolated remote control signal. For example, the zero-cross signal (a pulse synchronous with the ac input voltage passing through zero at every line cycle) could be utilized as an isolated signal for some electric appliances, such as for example washing machines to sense line frequency or generate timing signals necessary for an efficient load switching in the appliance.

(68) As shown in the illustrated example, the lead frame communication link **640** is bidirectional and includes two unidirectional communication links **641** and **642**. Communication link **642** is unidirectional in a reverse direction of lead frame communication link **641**. It is appreciated that even though in the illustrated example the individual lead frame communication links are described as unidirectional communication links, in another example, a single lead frame communication link can be utilized in a bidirectional implementation (as presented, for example, in FIG. 1B) instead of two unidirectional communication links in accordance with the teachings of the present invention.

(69) Even though the magnetically coupled communication link built in accordance with the teachings of this invention provides galvanic isolation, one skilled in the art would appreciate that the overall system, such as a switch mode power converter, that practices this invention need not necessarily be galvanically isolated between the input and output of the system to benefit from the advantages of this invention. For example, in non-isolated converters, a galvanically isolated communication link in accordance with the teachings of the present invention allows communication between two parts of a switch mode power converter that are referenced to different voltages which could be fixed or changing relative each other over time.

(70) The above description of illustrated examples of the present invention, including what is described in the Abstract, are not intended to be exhaustive or to be limitation to the precise forms disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible without departing from the broader spirit and scope of the present invention. Indeed, it is appreciated that the specific example voltages, currents, frequencies, power range values, times, etc., are provided for explanation

purposes and that other values may also be employed in other embodiments and examples in accordance with the teachings of the present invention.

## Claims

1. A lead frame comprising: a first conductor comprising an inner conductive loop, and a second conductor galvanically isolated from the first conductor, wherein the second conductor comprises an outer conductive loop disposed proximate to and magnetically coupled to the inner conductive loop to provide a communication link between the first and second conductors, wherein substantially all of the outer conductive loop is outside of the inner conductive loop; a first ground pad coupled to the first conductor; and a second ground pad comprising a slot formed by an inner portion and an outer portion coupled to the second conductor such that the outer portion extends the outer conductive loop.
  2. The lead frame of claim 1 wherein the magnetically coupled portion of the inner and outer conductive loops are disposed substantially in a same plane.
  3. The lead frame of claim 2 wherein the inner and outer conductive loops each consist of one turn.
  4. The lead frame of claim 2, wherein the inner conductive loop and the outer conductive loop are physically symmetric.
  5. The lead frame of claim 2, comprising: a tie bar connection; and the first conductor further comprising a third conductive loop coupled to the tie bar connection and configured for noise cancellation.
  6. A lead frame comprising: a first conductor comprising an inner conductive loop; and a second conductor galvanically isolated from the first conductor, wherein the second conductor comprises an outer conductive loop disposed proximate to and magnetically coupled to the inner conductive loop to provide a communication link between the first and second conductors, and wherein substantially all of the outer conductive loop is outside of the inner conductive loop.
  7. The lead frame as in claim 6 wherein the inner conductive loop and the outer conductive loop are concentric.
  8. The lead frame of claim 7 comprising: a first ground pad coupled to the first conductor; and a second ground pad comprising a slot formed by an inner portion and an outer portion coupled to the second conductor such that the outer portion extends the outer conductive loop.
  9. The lead frame of claim 8 wherein the magnetically coupled portion of the inner and outer conductive loops are disposed substantially in a same plane.
  10. The lead frame of claim 9 wherein the inner and outer conductive loops each consist of one turn.
  11. The lead frame of claim 9, wherein the inner conductive loop and the outer conductive loop are physically symmetric.
  12. The lead frame of claim 9 comprising: a tie bar connection; and the first conductor further comprising a third conductive loop coupled to the tie bar connection and configured for noise cancellation.
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